

NKPC HSA Bayesian State-Space Estimation: Output gap, BN filter

June 22, 2026

1 Data and Transformations

Raw data are read from `data/raw/`. Processed model-ready data are written to `data/processed/`. The HSA competition aggregator convention is $N_t = 100 \log(N_t^{level})$ unless a run metadata file explicitly states that the series was already transformed.

2 Model Specifications

The CES benchmark is

$$\pi_t = \alpha\pi_{t-1} + (1 - \alpha)E_t\pi_{t+1} + \kappa x_t + e_t.$$

The dynamic HSA specification is

$$\pi_t = \alpha\pi_{t-1} + (1 - \alpha)E_t\pi_{t+1} + \kappa x_t - \theta\hat{N}_t + e_t.$$

The steady HSA specification is

$$\pi_t = \alpha\pi_{t-1} + (1 - \alpha)E_t\pi_{t+1} + (\kappa_0 + \delta\bar{N}_t)x_t + e_t.$$

The both-effects HSA specification is

$$\pi_t = \alpha\pi_{t-1} + (1 - \alpha)E_t\pi_{t+1} + (\kappa_0 + \delta\bar{N}_t)x_t - (\theta_0 + \gamma\bar{N}_t)\hat{N}_t + e_t.$$

The maintained HSA decomposition is

$$N_t = \hat{N}_t + \bar{N}_t, \quad \hat{N}_t = \rho_1\hat{N}_{t-1} + \rho_2\hat{N}_{t-2} + u_t, \quad \bar{N}_t = n + \bar{N}_{t-1} + \epsilon_t.$$

3 Unit and Covariance Conventions

Priors for κ , κ_0 , and δ are specified in physical units. Legacy Gibbs samplers internally use `KAPPA.SCALE=100` when the design matrix contains $x_t/100$. All posterior draws, tables, figures, SDDR calculations, and model-comparison summaries are reported in physical units. The default dynamic-HSA shock covariance allows only (e_t, ζ_t) correlation; diagonal and historical full covariance modes are available only through explicit run metadata.

4 Prior Specifications

Baseline, weak, and tight priors are stored in `configs/`. Robustness tables below summarize whether conclusions for δ , θ , and γ change across these choices.

5 1. Posterior Mean Coefficients

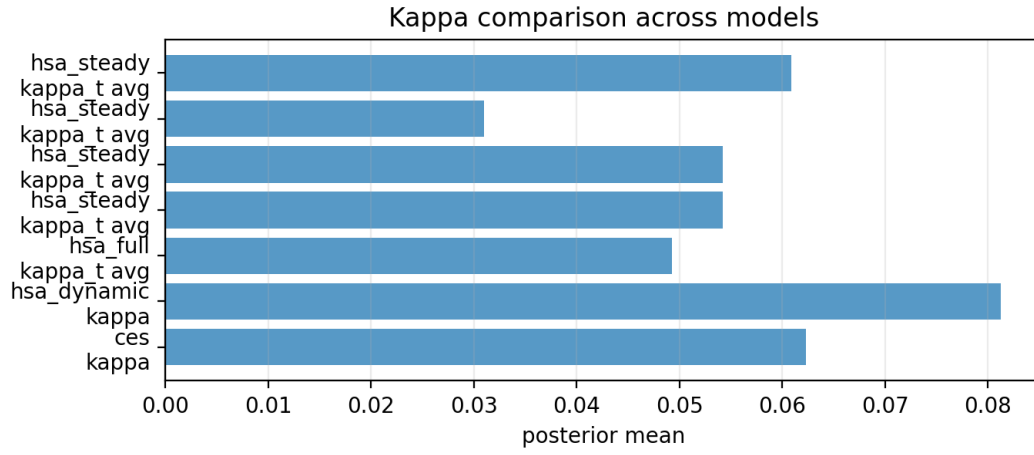
This section reports posterior means and credible intervals for the main scalar coefficients in each estimated run.

run	model	parameter	posterior_mean	posterior_sd	ci_2.5	ci_97.5
ces_output_gap_bn_baseline_20260622.131732	ces	alpha	0.6692	0.0692	0.5437	0.7835
ces_output_gap_bn_baseline_20260622.131732	ces	kappa	0.0623	0.0493	-0.0059	0.1764
ces_output_gap_bn_baseline_20260622.131732	ces	phi.1	0.9192	0.0369	0.8378	0.9703
ces_output_gap_bn_baseline_20260622.131732	ces	lambda_ez	-0.0085	0.1099	-0.2047	0.2214
hsa_dynamic_output_gap_bn_baseline_20260622.131735	hsa_dynamic	alpha	0.6417	0.0572	0.5407	0.7302
hsa_dynamic_output_gap_bn_baseline_20260622.131735	hsa_dynamic	kappa	0.0812	0.0406	-0.0037	0.1503
hsa_dynamic_output_gap_bn_baseline_20260622.131735	hsa_dynamic	theta	0.0089	0.0035	0.0011	0.0144
hsa_dynamic_output_gap_bn_baseline_20260622.131735	hsa_dynamic	phi.1	0.9069	0.0263	0.8613	0.9529
hsa_dynamic_output_gap_bn_baseline_20260622.131735	hsa_dynamic	rho.1	1.8622	0.0360	1.7937	1.9130
hsa_dynamic_output_gap_bn_baseline_20260622.131735	hsa_dynamic	rho.2	-0.8657	0.0358	-0.9167	-0.7966
hsa_dynamic_output_gap_bn_baseline_20260622.131735	hsa_dynamic	n	-0.0263	0.0733	-0.1502	0.1060
hsa_full_output_gap_bn_baseline_20260622.131737	hsa_full	alpha	0.6624	0.0547	0.5625	0.7575
hsa_full_output_gap_bn_baseline_20260622.131737	hsa_full	kappa_0	0.2476	0.1843	-0.0392	0.5474
hsa_full_output_gap_bn_baseline_20260622.131737	hsa_full	delta	-0.0010	0.0010	-0.0023	0.0005
hsa_full_output_gap_bn_baseline_20260622.131737	hsa_full	theta_0	0.1009	0.1795	-0.1749	0.4086
hsa_full_output_gap_bn_baseline_20260622.131737	hsa_full	gamma	-0.0003	0.0009	-0.0017	0.0012
hsa_full_output_gap_bn_baseline_20260622.131737	hsa_full	phi.1	0.9163	0.0287	0.8724	0.9763
hsa_full_output_gap_bn_baseline_20260622.131737	hsa_full	rho.1	1.1912	0.1128	1.0032	1.3882
hsa_full_output_gap_bn_baseline_20260622.131737	hsa_full	rho.2	-0.2689	0.1196	-0.4702	-0.0667
hsa_full_output_gap_bn_baseline_20260622.131737	hsa_full	n	-0.0899	0.0590	-0.1806	0.0369
hsa_full_output_gap_bn_baseline_20260622.131737	hsa_full	lambda_ez	-0.0133	0.1240	-0.1946	0.2723
hsa_steady_output_gap_bn_baseline_20260622.131733	hsa_steady	alpha	0.6552	0.0662	0.5340	0.7545
hsa_steady_output_gap_bn_baseline_20260622.131733	hsa_steady	kappa_0	0.1364	0.1960	-0.2435	0.4295
hsa_steady_output_gap_bn_baseline_20260622.131733	hsa_steady	delta	-0.0004	0.0010	-0.0017	0.0016
hsa_steady_output_gap_bn_baseline_20260622.131733	hsa_steady	phi.1	0.9184	0.0313	0.8669	0.9740
hsa_steady_output_gap_bn_baseline_20260622.131733	hsa_steady	rho.1	1.8025	0.0502	1.7002	1.8931
hsa_steady_output_gap_bn_baseline_20260622.131733	hsa_steady	rho.2	-0.8061	0.0505	-0.8996	-0.7028
hsa_steady_output_gap_bn_baseline_20260622.131733	hsa_steady	n	-0.0567	0.0737	-0.1998	0.0588
hsa_steady_output_gap_bn_baseline_20260622.131733	hsa_steady	lambda_ez	0.0007	0.0960	-0.1614	0.1608
hsa_steady_output_gap_bn_baseline_20260622.132042	hsa_steady	alpha	0.6552	0.0662	0.5340	0.7545
hsa_steady_output_gap_bn_baseline_20260622.132042	hsa_steady	kappa_0	0.1364	0.1960	-0.2435	0.4295
hsa_steady_output_gap_bn_baseline_20260622.132042	hsa_steady	delta	-0.0004	0.0010	-0.0017	0.0016
hsa_steady_output_gap_bn_baseline_20260622.132042	hsa_steady	phi.1	0.9184	0.0313	0.8669	0.9740
hsa_steady_output_gap_bn_baseline_20260622.132042	hsa_steady	rho.1	1.8025	0.0502	1.7002	1.8931
hsa_steady_output_gap_bn_baseline_20260622.132042	hsa_steady	rho.2	-0.8061	0.0505	-0.8996	-0.7028
hsa_steady_output_gap_bn_baseline_20260622.132042	hsa_steady	n	-0.0567	0.0737	-0.1998	0.0588
hsa_steady_output_gap_bn_baseline_20260622.132042	hsa_steady	lambda_ez	0.0007	0.0960	-0.1614	0.1608
hsa_steady_output_gap_bn_tight_20260622.132047	hsa_steady	alpha	0.6354	0.0644	0.5449	0.7738
hsa_steady_output_gap_bn_tight_20260622.132047	hsa_steady	kappa_0	0.1386	0.1067	-0.0929	0.3230
hsa_steady_output_gap_bn_tight_20260622.132047	hsa_steady	delta	-0.0005	0.0007	-0.0020	0.0009
hsa_steady_output_gap_bn_tight_20260622.132047	hsa_steady	phi.1	0.8995	0.0282	0.8447	0.9392
hsa_steady_output_gap_bn_tight_20260622.132047	hsa_steady	rho.1	0.5635	0.1054	0.3515	0.7331
hsa_steady_output_gap_bn_tight_20260622.132047	hsa_steady	rho.2	-0.3716	0.1139	-0.5634	-0.1618
hsa_steady_output_gap_bn_tight_20260622.132047	hsa_steady	n	-0.0197	0.0444	-0.0969	0.0550
hsa_steady_output_gap_bn_tight_20260622.132047	hsa_steady	lambda_ez	-0.0094	0.1036	-0.2054	0.1170
hsa_steady_output_gap_bn_weak_20260622.132044	hsa_steady	alpha	0.6744	0.0603	0.5567	0.7893
hsa_steady_output_gap_bn_weak_20260622.132044	hsa_steady	kappa_0	0.3115	0.4374	-0.3883	1.0701
hsa_steady_output_gap_bn_weak_20260622.132044	hsa_steady	delta	-0.0013	0.0023	-0.0054	0.0028
hsa_steady_output_gap_bn_weak_20260622.132044	hsa_steady	phi.1	0.9201	0.0315	0.8757	0.9852
hsa_steady_output_gap_bn_weak_20260622.132044	hsa_steady	rho.1	1.9131	0.0382	1.8481	1.9768
hsa_steady_output_gap_bn_weak_20260622.132044	hsa_steady	rho.2	-0.9172	0.0385	-0.9796	-0.8540
hsa_steady_output_gap_bn_weak_20260622.132044	hsa_steady	n	-0.1524	0.1028	-0.3366	0.0072
hsa_steady_output_gap_bn_weak_20260622.132044	hsa_steady	lambda_ez	0.0088	0.0864	-0.1303	0.1975

6 2. Kappa Comparison Across Models

For CES models this is κ . For steady or full HSA models the table also reports κ_0 , δ , and the average path of $\kappa_t = \kappa_0 + \delta \bar{N}_t$ when available.

run	model	kappa_mean	kappa_ci.2.5	kappa_ci.97.5	kappa_0_mean	kappa_0_ci.2.5	kappa_0_ci.97.5	delta_mean	delta_ci.2.5	delta_ci.97.5	kappa_t_averall_mean	kappa_t_start_mean	kappa_t_end_mean
ces_output_gap_bn_baseline_20260622.131732	ces	0.0623	-0.0059	0.1764	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_output_gap_bn_baseline_20260622.131735	hsa_dynamic	0.0812	-0.0037	0.1503	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap_bn_baseline_20260622.131737	hsa_full	NaN	NaN	NaN	0.2476	-0.0392	0.5474	-0.0010	-0.0023	0.0005	0.0493	0.0531	0.0745
hsa_steady_output_gap_bn_baseline_20260622.131733	hsa_steady	NaN	NaN	NaN	0.1364	-0.2435	0.4295	-0.0004	-0.0017	0.0016	0.0542	0.0522	0.0567
hsa_steady_output_gap_bn_baseline_20260622.132042	hsa_steady	NaN	NaN	NaN	0.1364	-0.2435	0.4295	-0.0004	-0.0017	0.0016	0.0542	0.0522	0.0567
hsa_steady_output_gap_bn_tight_20260622.132047	hsa_steady	NaN	NaN	NaN	0.1386	-0.0929	0.3230	-0.0005	-0.0020	0.0009	0.0310	0.0322	0.0451
hsa_steady_output_gap_bn_weak_20260622.132044	hsa_steady	NaN	NaN	NaN	0.3115	-0.3883	1.0701	-0.0013	-0.0054	0.0028	0.0608	0.0449	0.0770

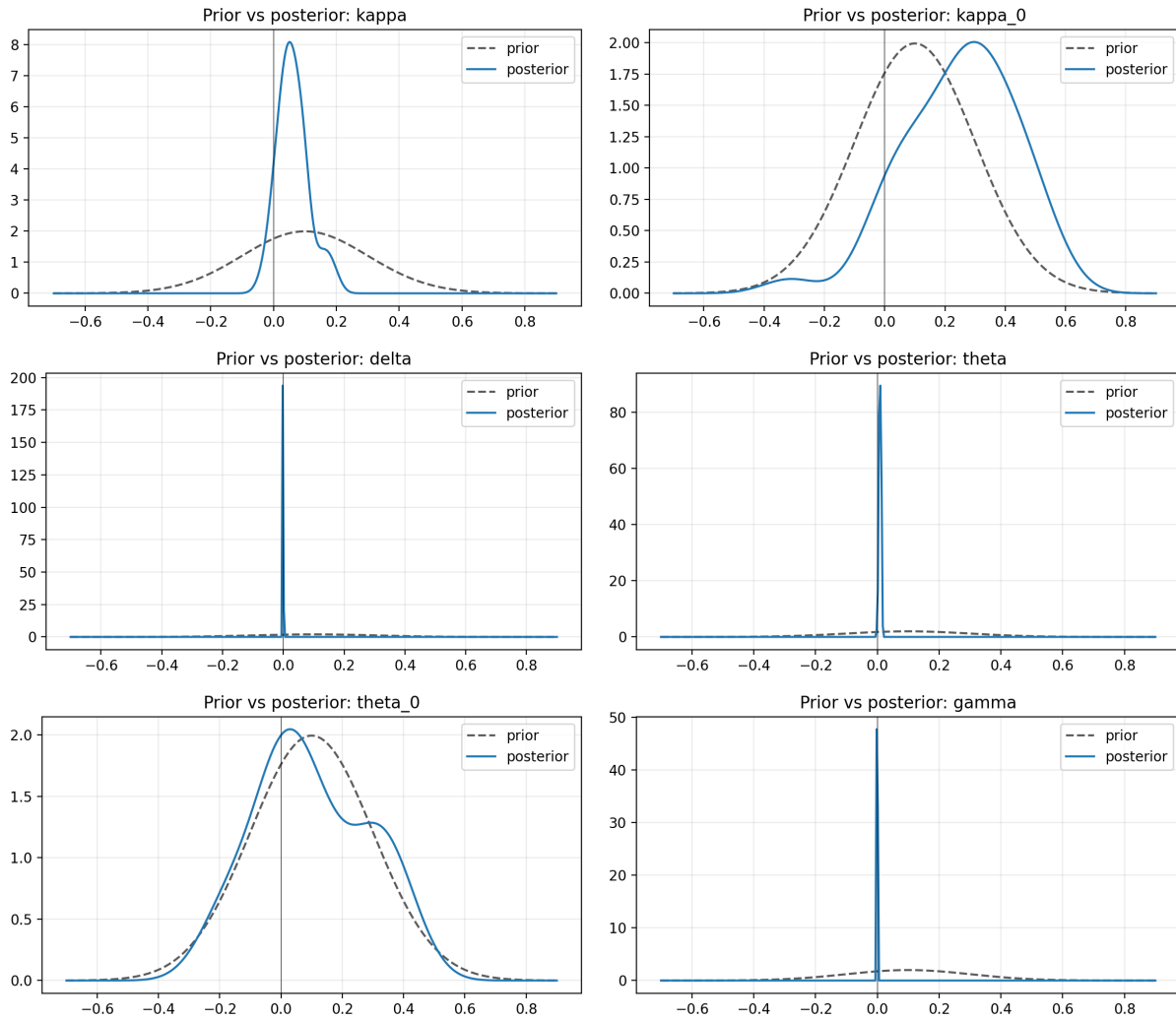


7 3. Time-Varying Coefficient Paths

The main time-varying coefficients are κ_t and, in the both-effects model, $\theta_t = \theta_0 + \gamma \bar{N}_t$.

run	model	coefficient	time_index	posterior_mean	ci.2.5	ci.97.5
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	0	0.0531	-0.0205	0.1427
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	5	0.0524	-0.0247	0.1445
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	11	0.0473	-0.0336	0.1347
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	16	0.0380	-0.0469	0.1201
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	21	0.0352	-0.0518	0.1265
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	27	0.0361	-0.0496	0.1224
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	32	0.0369	-0.0497	0.1236
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	37	0.0376	-0.0476	0.1223
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	43	0.0351	-0.0516	0.1262
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	48	0.0321	-0.0560	0.1233
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	53	0.0303	-0.0591	0.1204
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	59	0.0301	-0.0601	0.1179
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	64	0.0349	-0.0515	0.1246
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	70	0.0436	-0.0397	0.1272
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	75	0.0506	-0.0280	0.1383
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	80	0.0574	-0.0195	0.1478
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	86	0.0610	-0.0132	0.1544
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	91	0.0623	-0.0122	0.1588
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	96	0.0635	-0.0098	0.1591
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	102	0.0669	-0.0071	0.1642
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	107	0.0709	-0.0055	0.1695
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	112	0.0713	-0.0049	0.1696
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	118	0.0740	-0.0043	0.1732
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa.t	123	0.0745	-0.0040	0.1733
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	0	0.0493	0.0052	0.0959
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	5	0.0491	0.0073	0.0959
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	11	0.0478	0.0098	0.0903
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	16	0.0454	0.0019	0.0922
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	21	0.0446	0.0007	0.0957
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	27	0.0448	0.0015	0.0936
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	32	0.0450	0.0020	0.0923
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	37	0.0452	0.0019	0.0925
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	43	0.0447	0.0003	0.0948
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	48	0.0436	-0.0053	0.0975
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	53	0.0431	-0.0064	0.1008
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	59	0.0430	-0.0102	0.0996
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	64	0.0443	0.0004	0.0960
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	70	0.0468	0.0084	0.0887
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	75	0.0486	0.0088	0.0958
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	80	0.0505	0.0018	0.1021
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	86	0.0515	-0.0014	0.1041
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	91	0.0521	-0.0031	0.1040
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	96	0.0522	-0.0030	0.1047
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	102	0.0531	-0.0039	0.1092
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	107	0.0542	-0.0045	0.1137
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	112	0.0541	-0.0046	0.1130
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	118	0.0549	-0.0050	0.1151
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta.t	123	0.0551	-0.0074	0.1172
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	0	0.0522	-0.0233	0.1296
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	5	0.0522	-0.0235	0.1306
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	11	0.0523	-0.0226	0.1298
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	16	0.0522	-0.0216	0.1290
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	21	0.0524	-0.0208	0.1288
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	27	0.0527	-0.0175	0.1317
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	32	0.0528	-0.0175	0.1319
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	37	0.0531	-0.0165	0.1320
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	43	0.0535	-0.0161	0.1325
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	48	0.0537	-0.0162	0.1326
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	53	0.0538	-0.0161	0.1332
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	59	0.0540	-0.0159	0.1323
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	64	0.0542	-0.0153	0.1325
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	70	0.0545	-0.0125	0.1324
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	75	0.0545	-0.0112	0.1325
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	80	0.0549	-0.0100	0.1326
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	86	0.0550	-0.0098	0.1326
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	91	0.0554	-0.0098	0.1331
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	96	0.0555	-0.0096	0.1334
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	102	0.0560	-0.0094	0.1331
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	107	0.0562	-0.0095	0.1324
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	112	0.0565	-0.0091	0.1329
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	118	0.0566	-0.0093	0.1335
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa.t	123	0.0567	-0.0093	0.1328
hsa_steady_output_gap_bn_baseline.20260622.132042	hsa_steady	kappa.t	0	0.0522	-0.0233	0.1296
hsa_steady_output_gap_bn_baseline.20260622.132042	hsa_steady	kappa.t	5	0.0522	-0.0235	0.1306
hsa_steady_output_gap_bn_baseline.20260622.132042	hsa_steady	kappa.t	11	0.0523	-0.0226	0.1298
hsa_steady_output_gap_bn_baseline.20260622.132042	hsa_steady	kappa.t	16	0.0522	-0.0216	0.1290
hsa_steady_output_gap_bn_baseline.20260622.132042	hsa_steady	kappa.t	21	0.0524	-0.0208	0.1288
hsa_steady_output_gap_bn_baseline.20260622.132042	hsa_steady	kappa.t	27	0.0527	-0.0175	0.1317
hsa_steady_output_gap_bn_baseline.20260622.132042	hsa_steady	kappa.t	32	0.0528	-0.0175	0.1319
hsa_steady_output_gap_bn_baseline.20260622.132042	hsa_steady	kappa.t	37	0.0531	-0.0165	0.1320

run	model	restriction	prior_mean	prior_sd	sddr_bf01
ces_output_gap_bn_baseline.20260622.131732	ces	kappa=0	0.1000	0.2000	2.3727
hsa_dynamic_output_gap_bn_baseline.20260622.131735	hsa_dynamic	kappa=0	0.1000	0.2000	1.0740
hsa_dynamic_output_gap_bn_baseline.20260622.131735	hsa_dynamic	theta=0	0.1000	0.2000	6.4392
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	kappa_0=0	0.1000	0.2000	0.5349
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	delta=0	0.1000	0.2000	126.7357
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	theta_0=0	0.1000	0.2000	1.1348
hsa_full_output_gap_bn_baseline.20260622.131737	hsa_full	gamma=0	0.1000	0.2000	243.6010
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	kappa_0=0	0.1000	0.2000	0.6104
hsa_steady_output_gap_bn_baseline.20260622.131733	hsa_steady	delta=0	0.1000	0.2000	165.3272
hsa_steady_output_gap_bn_baseline.20260622.132042	hsa_steady	kappa_0=0	0.1000	0.2000	0.6104
hsa_steady_output_gap_bn_baseline.20260622.132042	hsa_steady	delta=0	0.1000	0.2000	165.3272
hsa_steady_output_gap_bn_tight.20260622.132047	hsa_steady	kappa_0=0	0.1000	0.2000	0.6489
hsa_steady_output_gap_bn_tight.20260622.132047	hsa_steady	delta=0	0.1000	0.2000	231.3719
hsa_steady_output_gap_bn_weak.20260622.132044	hsa_steady	kappa_0=0	0.1000	0.2000	0.3411
hsa_steady_output_gap_bn_weak.20260622.132044	hsa_steady	delta=0	0.1000	0.2000	68.8500



10 Diagnostics and Robustness

run	model	data_spec	prior_spec	parameter	mean	sd	hdi.2.5%	hdi.97.5%	mcmc.mean	ess_bulk	ess_tail	r_hat	warning
ces_output_gap.bn.baseline.20260622.131732	ces	output_gap.bn	baseline	alpha	0.6692	0.0692	0.5437	0.7835	NaN	NaN	NaN	NaN	NaN
ces_output_gap.bn.baseline.20260622.131732	ces	output_gap.bn	baseline	kappa	0.0623	0.0493	-0.0059	0.1764	NaN	NaN	NaN	NaN	NaN
ces_output_gap.bn.baseline.20260622.131732	ces	output_gap.bn	baseline	phi.1	0.9192	0.0369	0.8378	0.9703	NaN	NaN	NaN	NaN	NaN
ces_output_gap.bn.baseline.20260622.131732	ces	output_gap.bn	baseline	lambda.ez	-0.0085	0.1099	-0.2047	0.2214	NaN	NaN	NaN	NaN	NaN
ces_output_gap.bn.baseline.20260622.131732	ces	output_gap.bn	baseline	rho	-0.0101	0.0968	-0.1644	0.1836	NaN	NaN	NaN	NaN	NaN
ces_output_gap.bn.baseline.20260622.131732	ces	output_gap.bn	baseline	sigma.e	0.6579	0.0435	0.5851	0.7073	NaN	NaN	NaN	NaN	NaN
ces_output_gap.bn.baseline.20260622.131732	ces	output_gap.bn	baseline	sigma.zeta	0.5831	0.0355	0.5256	0.6426	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_output_gap.bn.baseline.20260622.131735	hsa_dynamic	output_gap.bn	baseline	alpha	0.6417	0.0572	0.5407	0.7302	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_output_gap.bn.baseline.20260622.131735	hsa_dynamic	output_gap.bn	baseline	kappa	0.0812	0.0406	-0.0037	0.1503	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_output_gap.bn.baseline.20260622.131735	hsa_dynamic	output_gap.bn	baseline	theta	0.0089	0.0035	0.0011	0.0144	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_output_gap.bn.baseline.20260622.131735	hsa_dynamic	output_gap.bn	baseline	phi.1	0.9069	0.0263	0.8613	0.9529	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_output_gap.bn.baseline.20260622.131735	hsa_dynamic	output_gap.bn	baseline	rho.1	1.8622	0.0360	1.7937	1.9130	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_output_gap.bn.baseline.20260622.131735	hsa_dynamic	output_gap.bn	baseline	rho.2	-0.8657	0.0358	-0.9167	-0.7966	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_output_gap.bn.baseline.20260622.131735	hsa_dynamic	output_gap.bn	baseline	n	-0.0263	0.0733	-0.1502	0.1060	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_output_gap.bn.baseline.20260622.131735	hsa_dynamic	output_gap.bn	baseline	sigma.e	0.6290	0.0351	0.5666	0.6908	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_output_gap.bn.baseline.20260622.131735	hsa_dynamic	output_gap.bn	baseline	sigma.zeta	0.5524	0.0337	0.4909	0.6128	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_output_gap.bn.baseline.20260622.131735	hsa_dynamic	output_gap.bn	baseline	sigma.u	0.4136	0.0376	0.3597	0.4857	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_output_gap.bn.baseline.20260622.131735	hsa_dynamic	output_gap.bn	baseline	sigma.eps	0.2394	0.0262	0.1906	0.2892	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_output_gap.bn.baseline.20260622.131735	hsa_dynamic	output_gap.bn	baseline	sigma.N	0.2806	0.0300	0.2338	0.3480	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap.bn.baseline.20260622.131737	hsa_full	output_gap.bn	baseline	alpha	0.6624	0.0547	0.5625	0.7575	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap.bn.baseline.20260622.131737	hsa_full	output_gap.bn	baseline	kappa.0	0.2476	0.1843	-0.0392	0.5474	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap.bn.baseline.20260622.131737	hsa_full	output_gap.bn	baseline	delta	-0.0010	0.0010	-0.0023	0.0005	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap.bn.baseline.20260622.131737	hsa_full	output_gap.bn	baseline	theta.0	0.1009	0.1795	-0.1749	0.4086	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap.bn.baseline.20260622.131737	hsa_full	output_gap.bn	baseline	gamma	-0.0003	0.0009	-0.0017	0.0012	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap.bn.baseline.20260622.131737	hsa_full	output_gap.bn	baseline	kappa.t.mean	0.0493	0.0481	-0.0296	0.1374	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap.bn.baseline.20260622.131737	hsa_full	output_gap.bn	baseline	kappa.t.start	0.0531	0.0467	-0.0205	0.1427	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap.bn.baseline.20260622.131737	hsa_full	output_gap.bn	baseline	kappa.t.mid	0.0324	0.0561	-0.0547	0.1230	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap.bn.baseline.20260622.131737	hsa_full	output_gap.bn	baseline	kappa.t.end	0.0745	0.0455	-0.0040	0.1733	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap.bn.baseline.20260622.131737	hsa_full	output_gap.bn	baseline	theta.t.mean	0.0483	0.0246	0.0089	0.0934	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap.bn.baseline.20260622.131737	hsa_full	output_gap.bn	baseline	theta.t.start	0.0493	0.0252	0.0052	0.0959	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap.bn.baseline.20260622.131737	hsa_full	output_gap.bn	baseline	theta.t.mid	0.0436	0.0281	-0.0046	0.0974	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap.bn.baseline.20260622.131737	hsa_full	output_gap.bn	baseline	theta.t.end	0.0551	0.0343	-0.0074	0.1172	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap.bn.baseline.20260622.131737	hsa_full	output_gap.bn	baseline	phi.1	0.9163	0.0287	0.8724	0.9763	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap.bn.baseline.20260622.131737	hsa_full	output_gap.bn	baseline	rho.1	1.1912	0.1128	1.0032	1.3882	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap.bn.baseline.20260622.131737	hsa_full	output_gap.bn	baseline	rho.2	-0.2689	0.1196	-0.4792	-0.0667	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap.bn.baseline.20260622.131737	hsa_full	output_gap.bn	baseline	n	-0.0899	0.0590	-0.1806	0.0369	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap.bn.baseline.20260622.131737	hsa_full	output_gap.bn	baseline	lambda.ez	-0.0133	0.1240	-0.1946	0.2723	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap.bn.baseline.20260622.131737	hsa_full	output_gap.bn	baseline	rho	-0.0103	0.1064	-0.1553	0.2447	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap.bn.baseline.20260622.131737	hsa_full	output_gap.bn	baseline	sigma.e	0.6709	0.0488	0.5815	0.7386	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap.bn.baseline.20260622.131737	hsa_full	output_gap.bn	baseline	sigma.zeta	0.5738	0.0376	0.5141	0.6553	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap.bn.baseline.20260622.131737	hsa_full	output_gap.bn	baseline	sigma.u	0.6961	0.0598	0.6001	0.8421	NaN	NaN	NaN	NaN	NaN
hsa_full_output_gap.bn.baseline.20260622.131737	hsa_full	output_gap.bn	baseline	sigma.eps	1.0609	0.0795	0.9293	1.1809	NaN	NaN	NaN	NaN	NaN
hsa_steady_output_gap.bn.baseline.20260622.131733	hsa_steady	output_gap.bn	baseline	alpha	0.6552	0.0662	0.5340	0.7545	NaN	NaN	NaN	NaN	NaN
hsa_steady_output_gap.bn.baseline.20260622.131733	hsa_steady	output_gap.bn	baseline	kappa.0	0.1364	0.1960	-0.2435	0.4295	NaN	NaN	NaN	NaN	NaN
hsa_steady_output_gap.bn.baseline.20260622.131733	hsa_steady	output_gap.bn	baseline	delta	-0.0004	0.0010	-0.0017	0.0016	NaN	NaN	NaN	NaN	NaN
hsa_steady_output_gap.bn.baseline.20260622.131733	hsa_steady	output_gap.bn	baseline	kappa.t.mean	0.0542	0.0423	-0.0126	0.1324	NaN	NaN	NaN	NaN	NaN
hsa_steady_output_gap.bn.baseline.20260622.131733	hsa_steady	output_gap.bn	baseline	kappa.t.start	0.0522	0.0424	-0.0233	0.1296	NaN	NaN	NaN	NaN	NaN
hsa_steady_output_gap.bn.baseline.20260622.131733	hsa_steady	output_gap.bn	baseline	kappa.t.mid	0.0542	0.0422	-0.0152	0.1324	NaN	NaN	NaN	NaN	NaN
hsa_steady_output_gap.bn.baseline.20260622.131733	hsa_steady	output_gap.bn	baseline	kappa.t.end	0.0567	0.0438	-0.0093	0.1328	NaN	NaN	NaN	NaN	NaN
hsa_steady_output_gap.bn.baseline.20260622.131733	hsa_steady	output_gap.bn	baseline	phi.1	0.9184	0.0313	0.8669	0.9740	NaN	NaN	NaN	NaN	NaN
hsa_steady_output_gap.bn.baseline.20260622.131733	hsa_steady	output_gap.bn	baseline	rho.1	1.8025	0.0502	1.7002	1.8931	NaN	NaN	NaN	NaN	NaN
hsa_steady_output_gap.bn.baseline.20260622.131733	hsa_steady	output_gap.bn	baseline	rho.2	-0.8061	0.0505	-0.8996	-0.7028	NaN	NaN	NaN	NaN	NaN
hsa_steady_output_gap.bn.baseline.20260622.131733	hsa_steady	output_gap.bn	baseline	n	-0.0567	0.0737	-0.1998	0.0588	NaN	NaN	NaN	NaN	NaN
hsa_steady_output_gap.bn.baseline.20260622.131733	hsa_steady	output_gap.bn	baseline	lambda.ez	0.0007	0.0960	-0.1614	0.1608	NaN	NaN	NaN	NaN	NaN
hsa_steady_output_gap.bn.baseline.20260622.131733	hsa_steady	output_gap.bn	baseline	rho	0.0035	0.0841	-0.1299	0.1487	NaN	NaN	NaN	NaN	NaN
hsa_steady_output_gap.bn.baseline.20260622.131733	hsa_steady	output_gap.bn	baseline	sigma.eta	0.6659	0.0385	0.6072	0.7428	NaN	NaN	NaN	NaN	NaN
hsa_steady_output_gap.bn.baseline.20260622.131733	hsa_steady	output_gap.bn	baseline	sigma.e	0.6683	0.0392	0.6126	0.7439	NaN	NaN	NaN	NaN	NaN
hsa_steady_output_gap.bn.baseline.20260622.131733	hsa_steady	output_gap.bn	baseline	sigma.zeta	0.5837	0.0356	0.5209	0.6448	NaN	NaN	NaN	NaN	NaN
hsa_steady_output_gap.bn.baseline.20260622.131733	hsa_steady	output_gap.bn	baseline	sigma.u	0.5135	0.0601	0.4415	0.6337	NaN	NaN	NaN	NaN	NaN
hsa_steady_output_gap.bn.baseline.20260622.131733	hsa_steady	output_gap.bn	baseline	sigma.eps	0.3848	0.0475	0.3123	0.4833	NaN	NaN	NaN	NaN	NaN
hsa_steady_output_gap.bn.baseline.20260622.131733	hsa_steady	output_gap.bn	baseline	sigma.N	0.2996	0.0238	0.2596	0.3365	NaN	NaN	NaN	NaN	NaN

data_spec	prior	parameter	mean	median	ci_2.5	ci_97.5	p_gt_0	p_lt_0	conclusion
output_gap_bn	baseline	alpha	0.6552	0.6674	0.5340	0.7545	1.0000	0.0000	NaN
output_gap_bn	baseline	kappa_0	0.1364	0.1811	-0.2435	0.4295	0.8000	0.2000	NaN
output_gap_bn	baseline	delta	-0.0004	-0.0006	-0.0017	0.0016	0.2500	0.7500	NaN
output_gap_bn	baseline	rho_1	1.8025	1.7995	1.7002	1.8931	1.0000	0.0000	NaN
output_gap_bn	baseline	rho_2	-0.8061	-0.8033	-0.8996	-0.7028	0.0000	1.0000	NaN
output_gap_bn	baseline	sigma_e	0.6683	0.6673	0.6126	0.7439	1.0000	0.0000	NaN
output_gap_bn	baseline	sigma_zeta	0.5837	0.5794	0.5209	0.6448	1.0000	0.0000	NaN
output_gap_bn	baseline	sigma_u	0.5135	0.5022	0.4415	0.6337	1.0000	0.0000	NaN
output_gap_bn	baseline	sigma_eps	0.3848	0.3797	0.3123	0.4833	1.0000	0.0000	NaN
output_gap_bn	baseline	sigma_N	0.2996	0.3019	0.2596	0.3365	1.0000	0.0000	NaN
output_gap_bn	weak	alpha	0.6744	0.6902	0.5567	0.7893	1.0000	0.0000	NaN
output_gap_bn	weak	kappa_0	0.3115	0.3900	-0.3883	1.0701	0.7500	0.2500	NaN
output_gap_bn	weak	delta	-0.0013	-0.0016	-0.0054	0.0028	0.3000	0.7000	NaN
output_gap_bn	weak	rho_1	1.9131	1.9078	1.8481	1.9768	1.0000	0.0000	NaN
output_gap_bn	weak	rho_2	-0.9172	-0.9136	-0.9796	-0.8540	0.0000	1.0000	NaN
output_gap_bn	weak	sigma_e	0.6544	0.6444	0.5987	0.7424	1.0000	0.0000	NaN
output_gap_bn	weak	sigma_zeta	0.5774	0.5756	0.5283	0.6438	1.0000	0.0000	NaN
output_gap_bn	weak	sigma_u	0.4021	0.4096	0.3227	0.4756	1.0000	0.0000	NaN
output_gap_bn	weak	sigma_eps	0.2940	0.2877	0.2353	0.3980	1.0000	0.0000	NaN
output_gap_bn	weak	sigma_N	0.2335	0.2293	0.2019	0.2807	1.0000	0.0000	NaN
output_gap_bn	tight	alpha	0.6354	0.6330	0.5449	0.7738	1.0000	0.0000	NaN
output_gap_bn	tight	kappa_0	0.1386	0.1442	-0.0929	0.3230	0.8750	0.1250	NaN
output_gap_bn	tight	delta	-0.0005	-0.0005	-0.0020	0.0009	0.1750	0.8250	NaN
output_gap_bn	tight	rho_1	0.5635	0.5612	0.3515	0.7331	1.0000	0.0000	NaN
output_gap_bn	tight	rho_2	-0.3716	-0.3916	-0.5634	-0.1618	0.0000	1.0000	NaN
output_gap_bn	tight	sigma_e	0.6686	0.6644	0.6191	0.7473	1.0000	0.0000	NaN
output_gap_bn	tight	sigma_zeta	0.5664	0.5628	0.5151	0.6292	1.0000	0.0000	NaN
output_gap_bn	tight	sigma_u	0.5522	0.5539	0.4803	0.6443	1.0000	0.0000	NaN
output_gap_bn	tight	sigma_eps	1.2563	1.2557	1.1292	1.4280	1.0000	0.0000	NaN
output_gap_bn	tight	sigma_N	0.5393	0.5354	0.4411	0.6388	1.0000	0.0000	NaN
output_gap_bn	all	delta_robustness	NaN	NaN	NaN	NaN	NaN	NaN	robust

note

Run scripts/05_period_robustness.py first.

11 Posterior Predictive Checks and Figures

Posterior predictive checks are generated after model runs.

12 Notes

WAIC and LOO are not used as primary criteria because the models contain latent states. Chib marginal likelihoods are reported only when the legacy ordinate calculation can be matched to the sampler units and supplied data.